

Wireless LAN III

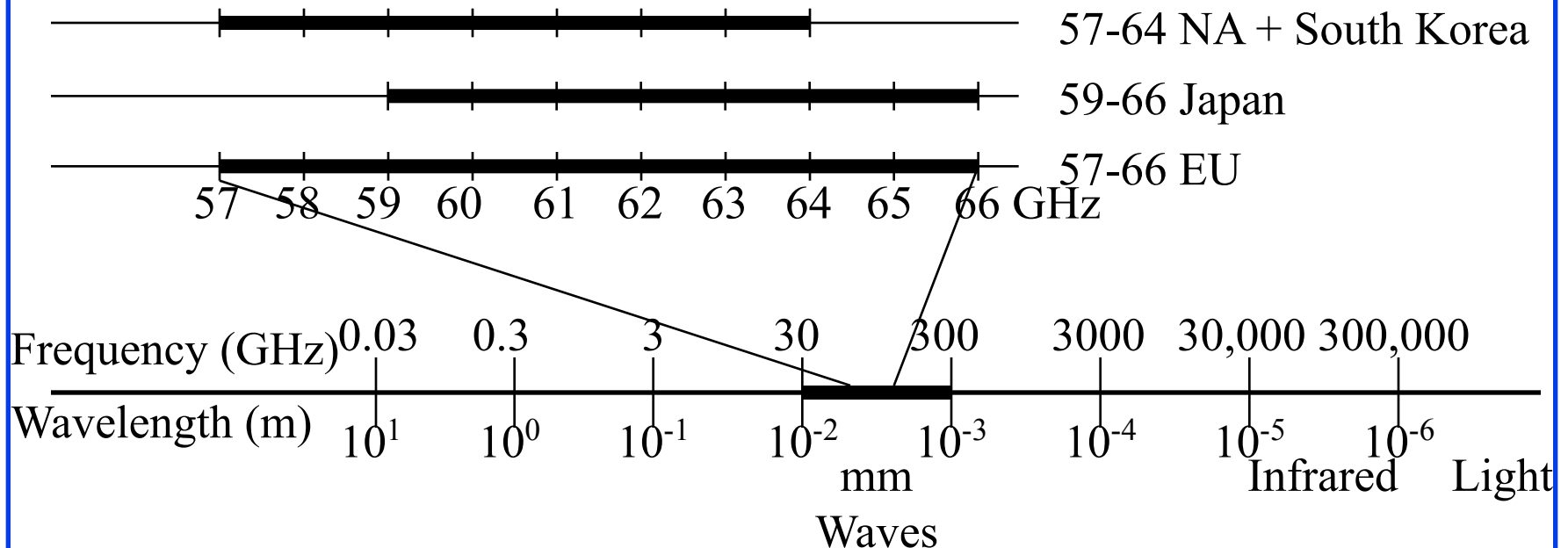
IEEE 802.11ad 60 GHz WLAN

Overview

1. **60 GHz Frequency Allocations and characteristics**
2. **60 GHz Wireless Standards**
3. **IEEE 802.11ad**
 - **Network topology and MAC**
 - **Beamforming**
 - **Clustering**
 - **Spatial Frequency Sharing**
 - **Relaying**

60 GHz Frequency Allocations

- 7-9 GHz in 57-66 GHz (**millimeter** waves 30GHz-300GHz)
- 4 Channels of ~ 2 GHz
- Significant activity after FCC made 57-64 GHz license-exempt



Ref: FCC, "Part 15 Rules for Unlicensed Operation in the 57-64 GHz Band," FCC13-112, August 2013,
http://hraunfoss.fcc.gov/edocs_public/attachmatch/FCC-13-112A1.pdf

Advantages of 60 GHz Band

1. **Large spectrum:** 7 GHz

- 7 Gbps requires only 1 b/Hz (BPSK ok).
- Complex 256-QAM not needed

2. **Small Antenna Separation:**

5 mm wavelength. $\lambda/4 = 1.25$ mm

3. **Easy Beamforming:** Antenna arrays on a chip.

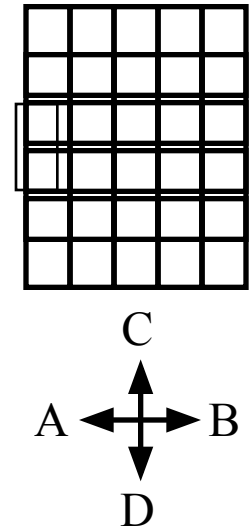
4. **Low Interference:** Does not cross walls.
Good for urban neighbors

5. **Directional Antennas:** Spatial reuse is easy

6. **Inherent security:** Difficult to intercept

7. **Higher power transmission:**

- FCC allows up to 27 dBm at 60 GHz but amplifiers difficult
- 60 GHz: 10 dBm + 30 dBi Antenna gain = 40 dBm EIRP
- 802.11n: 22 dBm + 3 dBi Antenna gain = 25 dBm EIRP



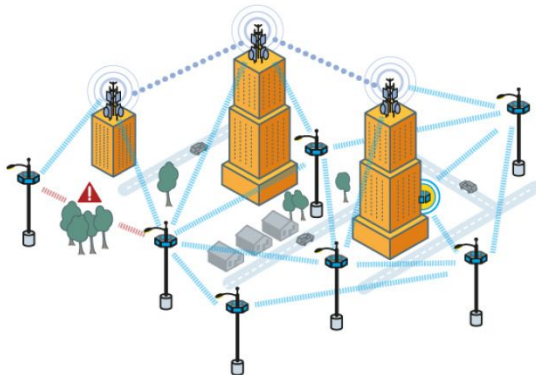
Disadvantages of 60 GHz Band

1. **Large Attenuation:** Attenuation / frequency²
 - ❑ Strong absorption by Oxygen
 - ❑ Need larger transmit power
 - ❑ Need high antenna gain $\Rightarrow$ directional antennas
 - ❑ Short Distance $\sim 10\text{m}$
2. **Directional Deafness:** Can't hear unless aligned
 - ❑ Carrier sense not possible
 - ❑ RTS/CTS does not work
 - ❑ Multicast Difficult
3. **Easily Blocked:** By a human/dog
Need a relay



Multi-Gigabit Wireless Applications

- ❑ **Cable Replacement:** High-Definition Uncompressed streaming video
- ❑ Interactive **gaming**
- ❑ High-speed file transfer
- ❑ Wireless Mesh **Backhaul** (200-400m)

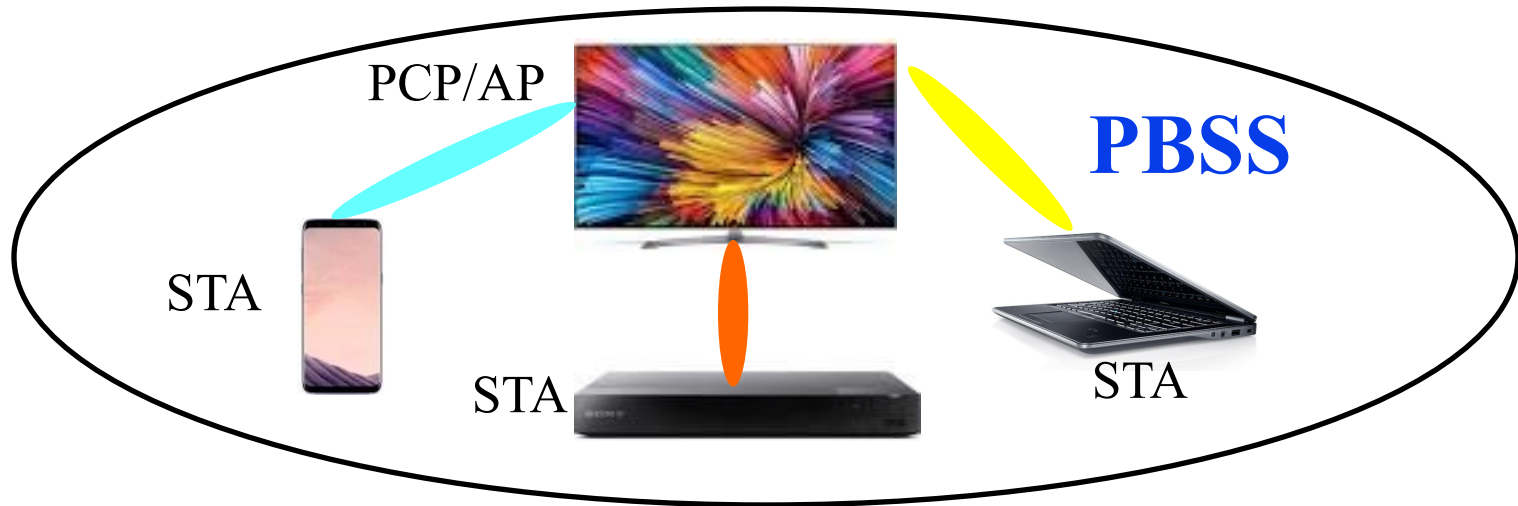


MAC Challenges for 60 GHz

- ❑ High path loss at 60 GHz
 - ❑ 28 dB higher than 2.4 GHz WLAN, 22 dB higher than 5 GHz WLAN
- ❑ Stations must have high antenna gain to overcome high path loss ❑
directional antenna/beamforming
 - ❑ The narrower the beam, the higher the antenna gains
- ❑ Directional communication complicates MAC design
 - ❑ AP can talk to a STA only if their beams point to each other
 - ❑ Similarly, two STAs must point their beams to each other before they can exchange data
- ❑ MAC must facilitate all stations to find the right beam directions at all times
 - ❑ Different beam directions for different destinations
 - ❑ Beam directions change with mobility
 - ❑ Becomes challenging with many stations in a network

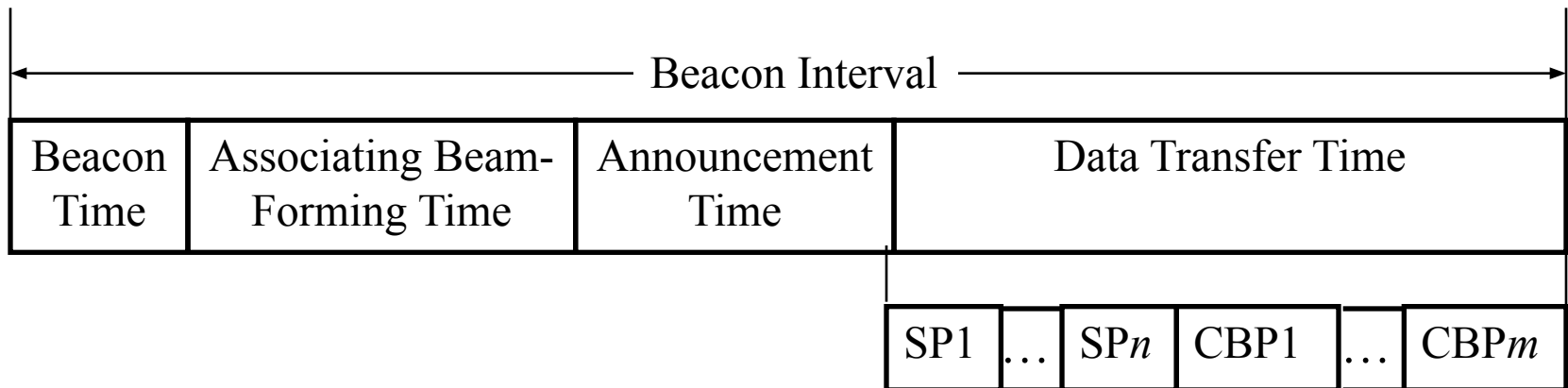
IEEE 802.11ad Network Topology

- ❑ **Personal Basic Service Set (PBSS):**
Group of stations that communicate
- ❑ **PBSS Central Point (PCP)** provides scheduling and timing using beacons
 - ❑ Dedicated AP not needed, PCP function can be assumed by any 802.11ad device, such as a TV in the room can be PCP/AP
- ❑ **1 PCP/AP per PBSS, 1-254 non-PCP stations (STA)**



802.11ad MAC – Beacon Interval

- Each super-frame called “**Beacon Interval**” is divided into 4 *access periods*: Beacon Time (**BT**), Associating Beamforming Training (**A-BFT**), Announcement Time (**AT**), and Data Transfer Time (**DTT**)

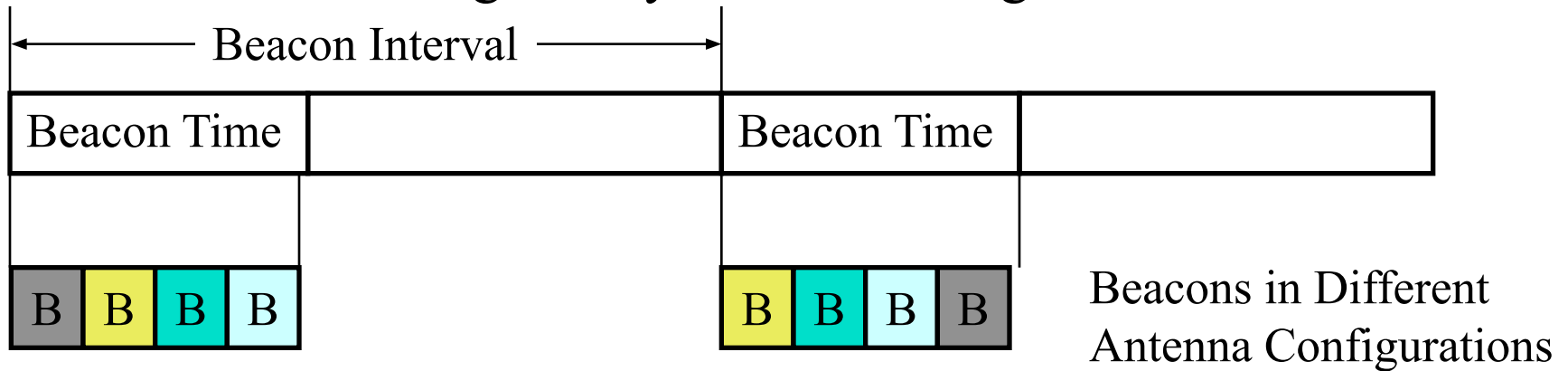


Access Periods in 802.11ad Beacon Interval

- ❑ **BT:** Only PCP can send a beacon during beacon time; PCP starts beamforming training in BT by sending training frames; STAs cannot transmit
- ❑ **A-BFT:** PCP performs antenna training with its members (STAs)
- ❑ **AT:** PCP polls members and receives non-data responses (STAs can request *service periods or SPs* to be scheduled during DTT)
- ❑ **DTT:** STA-to-STA exchange happens. All stations exchange data frames in a dedicated **service period (SP)** or by **contention in contention-based period (CBP)**
 - ❑ CBP uses Distributed Coordination Function (DCF); SP uses Hybrid Coordination Function (HCF)

IEEE 802.11ad Beacon

- Beacon transmissions are omni-directional $\Rightarrow$ One beacon is transmitted through every antenna configuration/sectors



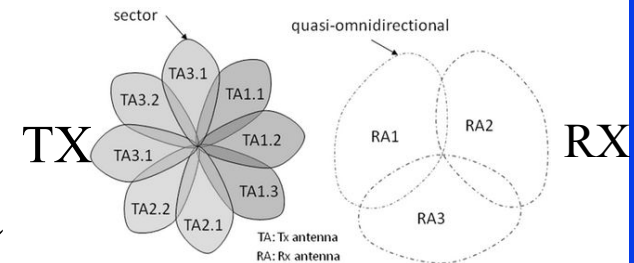
Example 1

- ❑ A 802.11ad PCP/AP has a four-sector antenna with every sector covering 90 degrees. During a Beacon Time (BT), how many beacons the AP should transmit?

Beacons must be transmitted in omni-direction, so all STAs receive it. 360 degree coverage is achieved by four sectors (each sector covers 90 degrees). The AP therefore will send four beacons, one per sector.

Beamforming Options

- ❑ The goal of beamforming is to choose the best beam direction pair that gives the highest gain
- ❑ Exhaustive search: exhaustively examine all beam pairs
 - ❑ E.g., 64 pairs for 8x8, 1024 pairs for 32x32
 - ❑ $O(B1 \times B2)$
 - ❑ Time and energy consuming



- ❑ Searching with omni-directional antenna
 - ❑ Transmitter transmits on all beam directions ($B1$ transmissions)
 - ❑ Receiver uses omni-directional antenna to record RSS for all beam directions of the transmitter
 - ❑ Receiver selects the highest RSS as the best beam for the transmitter
 - ❑ The pair takes turn to find each others best beam direction ($B2$ transmissions)
 - ❑ $O(B1+B2)$

Example 2

- ❑ Two 802.11ad devices, STA1 and STA2, want to beamform. STA1 has 32 different antenna configurations (i.e., capable of steering the beam to 32 different directions). STA2 has only 4 beam directions. For Exhaustive Search, how many training frames are transmitted in total by these two devices before they discover the optimum beam pairs for communication?

Total combinations of antenna configurations between the two stations is $32 \times 4 = 128$. Therefore, 128 training frames are transmitted, one per specific pair of antenna configurations, before the best combination (pair) is finally selected.

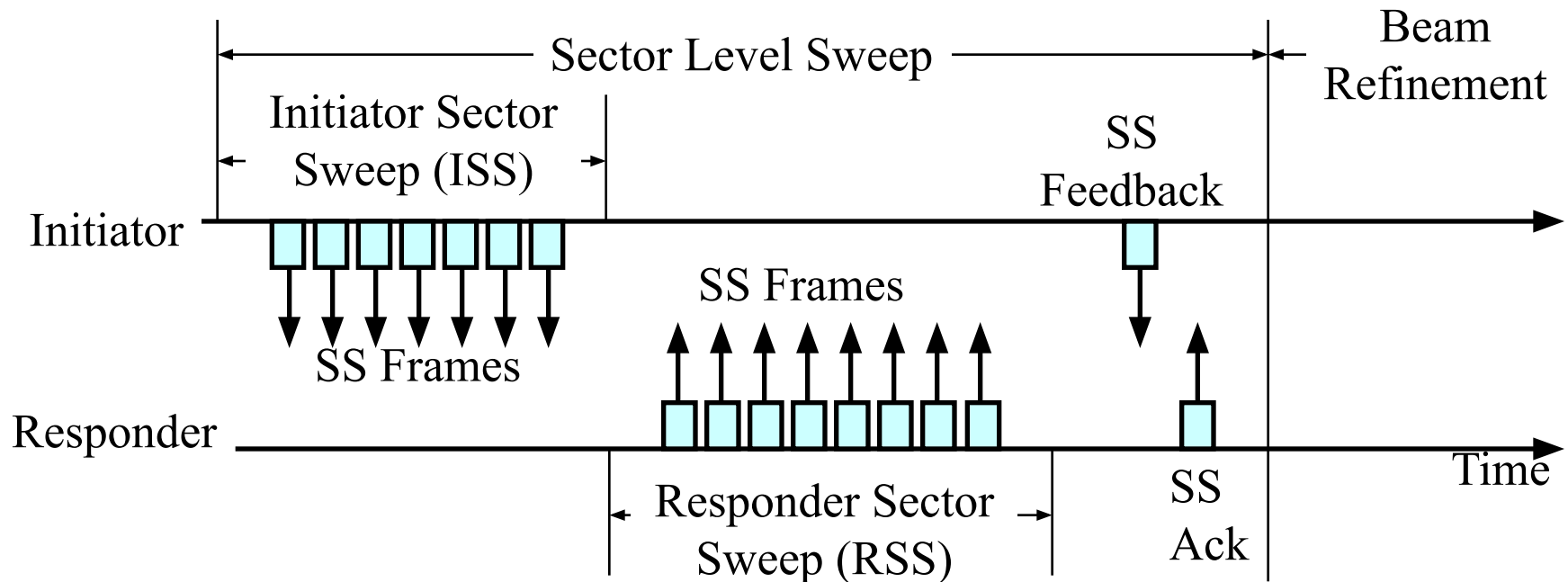
Example 3

- Two 802.11ad devices, STA1 and STA2, want to beamform. STA1 has 16 different antenna configurations (i.e., capable of steering the beam to 16 different directions). STA2 has only 4 beam directions. For Omni-direction Antenna approach, how many training frames are transmitted in total by these two devices before they discover the optimum beam pairs for communication?

STA1 first transmits 16 training frames while STA2 is listening in omni-direction. Then STA2 transmits 4 frames while STA2 is listening. Total frames transmitted = $16+4 = 20$.

IEEE 802.11ad Beamforming Training

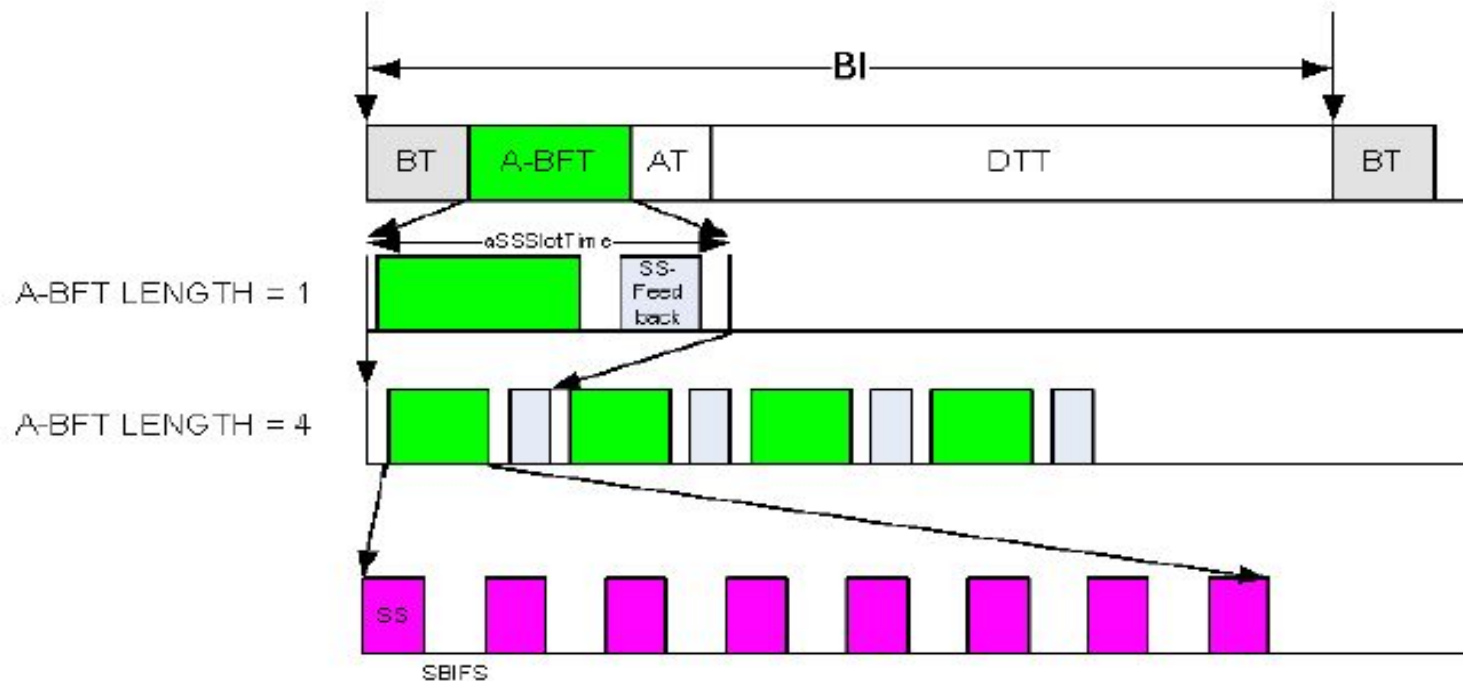
- ❑ Each station finds the optimal antenna configuration with its recipient using a two-stage search --- **SLS** followed by **BRP**
- ❑ **Sector Level Sweep (SLS):** Sends in all sectors and finds the optimal *sector pair*
 - ❑ Sector ❑ coarse direction (there can be several sharper beams within a sector)
 - ❑ Low data rate with sector-level beamforming
- ❑ **Beam Refinement Protocol (BRP):** Searches through the optimal sector to find the optimal parameters in that sector (identify a narrower beam)
 - ❑ Higher data rate (multi-gigabit) with beam refinement



AP-STA Beamforming

- ❑ Takes place during BT and A-BFT durations
- ❑ During BT, AP transmits training frames on all its sectors; all STAs listen in omni-direction mode
- ❑ A-BFT duration is slotted; each STA selects a slot randomly and transmits training frames on all its sectors; AP listens in omni-direction mode
- ❑ Random slot selection may lead to collision. No feedback from AP to the STA if collision; STA tries in next beacon interval
- ❑ Only SLS is completed in BT and A-BFT; BFP is optional and may take place in DT duration

SLS in A-BFT



Example: consider that the A-BFT LENGTH = 8 and 3 mSTAs A, B and C are competing. All mSTAs choose a random value between $[0,7]$. mSTA A chooses value = 2, while mSTAs B and C choose a value = 5 which may result in a collision.



Akhtar and Ergen, "Efficient Network Level Beamforming Training for IEEE 802.11ad WLANs", SPECTS 2015

Example 4

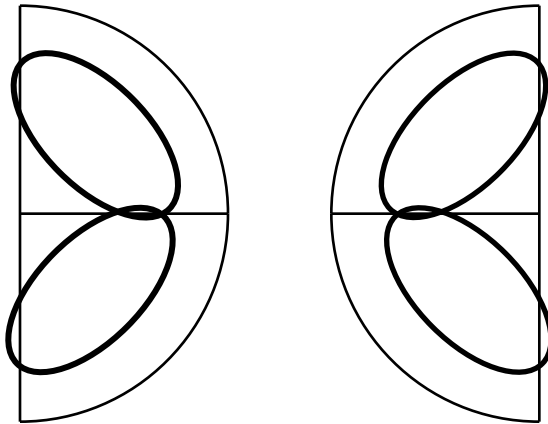
- The table shows the received signal strength (RSS) at the responder for each transmitted training frame from the beam training initiator during SLS. There are four sectors for both initiator and responder, and the number after the station letter denotes the sector number. For example, row 1 shows the frame transmitted by station A on its sector 1. What is the optimum beam (sector) pair discovered after the SLS?

The sector that produces the strongest signal is selected as the best sector. For A, the strongest sector is 3 (-50 dBm). For B, sector 1 produces the strongest signal at A (-49 dBm). The optimum beam pair for (A,B) therefore is (3,1).

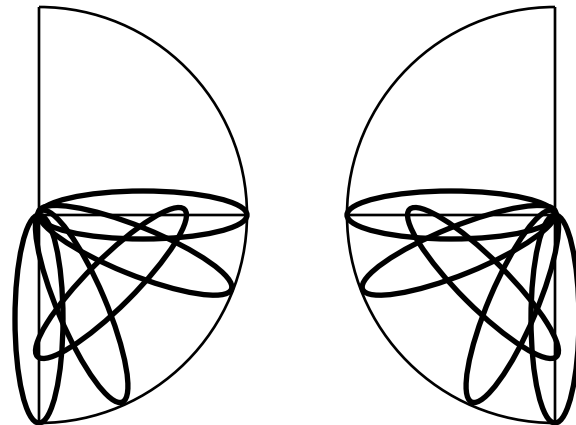
Transmitted Training Frame	RSS at Responder
A.1	-70 dBm
A.2	-62 dBm
A.3	-50 dBm
A.4	-64 dBm
B.1	-49 dBm
B.2	-71 dBm
B.3	-75 dBm
B.4	-80 dBm

SLS vs. Beam Refinement

- ❑ SLS uses “coarse” (wider) beams
 - ❑ Quick, but low gain and hence low data rate
- ❑ Beam refinement finds narrower beams within the same sector
 - ❑ Increases gain and data rate, but takes additional time before communication can begin



Sector-Level Sweep



Beam Refinement

Spatial Frequency Sharing (SFS)

- ❑ Multiple transmissions may be scheduled on the same frequency at the same time if they don't interfere
- ❑ PCP asks stations to send results of any STA-STA beamforming training outcomes. PCP then has the complete knowledge of beam pairing among the stations within its PBSS. PCP then can work out which station pairs can share the same slot (will not interfere).

Example 6

- In a given PBSS, all stations have 12 antenna sectors with 30 degree transmission angle. The table shows the beam pairs learned from beam training among 6 stations, A to F. For example, the first row of the table shows that A would use its beam #1 to communicate with B while B would use its beam #7 to communicate with A. If a communication, SP1, between A and B has already been scheduled, can SP2, a new communication between E and F, be spatially shared with SP1, i.e., be allocated during the same time slots without interference?

No. During SP1, B will transmit on its beam #7, which is the same beam number found to be optimum to communicate with E (Row 3 in the table). Therefore B's transmissions to A during SP1 will affect E. SP2 therefore cannot be spatially shared with SP1 without interference.

STA Pair	Beam Pair
(A,B)	(1,7)
(A,E)	(4,12)
(B,E)	(7,2)
(B,F)	(9,10)
(C,D)	(10,4)
(A,F)	(2,7)
(E,F)	(3,7)

Summary

1. 60 GHz, a.k.a. mm wave, has large bandwidth, small antenna separation allows easy beamforming and gigabit speeds but short distance due to large attenuation
2. Tri-band Wireless LAN devices with 2.4 GHz, 5.8GHz, and 60GHz are coming
3. 802.11ad LAN uses a PBSS central control point (PCP)
4. In all cases antenna alignment and tracking is required.
5. **Centralized** scheduling. Only **PCP** can send beacons. It sends beacons in all sectors.
6. Superframe (**Beacon Interval**) consists of Beacon Time, Associating Beamforming Training, Announcement Time, and Data Transfer Time
7. Announcement time is used for collecting requests from STAs
8. Data transfer can be pre-allocated or by contention
9. **Antenna training** is a 2-phase process. Sector selection and beam refinement.
10. Multiple transmission can take place on the same frequency at the same time (**Spatial Frequency Sharing**).